

VPC TASK-1

1. Create VPC with 2 private and 2 public subnets.

Step 1: Create a VPC

- Name: **my-vpc-01**
- CIDR block: **172.168.0.0/24**
→ This gives you **256 IP addresses**
- VPC created successfully.

Step 2: Create Public Subnets

I divided the VPC's 256 IPs into 4 smaller blocks (each /26 has 64 IPs).

Public Subnet 1

- Name: **Public-subnet-01**
- CIDR: **172.168.0.0/26**
→ First 64 IPs
- AZ: **eu-north-1a**

Public Subnet 2

- Name: **Public-subnet-02**
- CIDR: **172.168.0.64/26**
→ Next 64 IPs
- AZ: **eu-north-1a**

Step 3: Create Private Subnets

Private Subnet 1

- Name: **Private-subnet-01**
- CIDR: **172.168.0.128/26**
→ Next 64 IPs
- AZ: **eu-north-1b**

Private Subnet 2

- Name: **Private-subnet-02**
- CIDR: **172.168.0.192/26**
→ Last 64 IPs

- **AZ: eu-north-1b**

Step 4: Review VPC Resource Map

You now have:

1 VPC (my-vpc-01)

4 subnets

- 2 Public (in AZ a)
- 2 Private (in AZ b)

All 4 subnets linked to one route table

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

- ☒ VPC only

- ☐ VPC and more

Name tag - optional

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

my-vpc

IPv4 CIDR block [Info](#)

- IPv4 CIDR manual input

- ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

- No IPv6 CIDR block

- ☐ IPAM-allocated IPv6 CIDR block

- ☐ Amazon-provided IPv6 CIDR block

- ☐ IPv6 CIDR owned by me

aws

Europe (Stockholm)

Srihari (299582146630)

Srihari

VPC > Subnets > Create subnet

0

?

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-0435a85785bdb58f9 (my-vpc)

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public-Subnet-1

The name can be up to 256 characters long.

Availability Zone

[Info](#)

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-Subnet-2
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-north-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways

Subnets (7) [Info](#)
Last updated less than a minute ago
Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Public-Subnet-2	subnet-0dfc720a9aa1b18b4	Available	vpc-0435a85785bdb58f9 my-...	Off	10.0.2.0/24
☐	Public-Subnet-1	subnet-050f61ea237fd1d9	Available	vpc-0435a85785bdb58f9 my-...	Off	10.0.1.0/24
☐	private-Subnet-2	subnet-0f6a4d4d00324bafa	Available	vpc-0435a85785bdb58f9 my-...	Off	10.0.5.0/24
☐	Private-Subnet-1	subnet-0e927b6727d4c5570	Available	vpc-0435a85785bdb58f9 my-...	Off	10.0.0.0/24

Resource map [Info](#) [Show all details](#)

VPC
Your AWS virtual network
my-vpc

Subnets (4)
Subnets within this VPC

eu-north-1a

- private-Subnet-2
- Public-Subnet-1

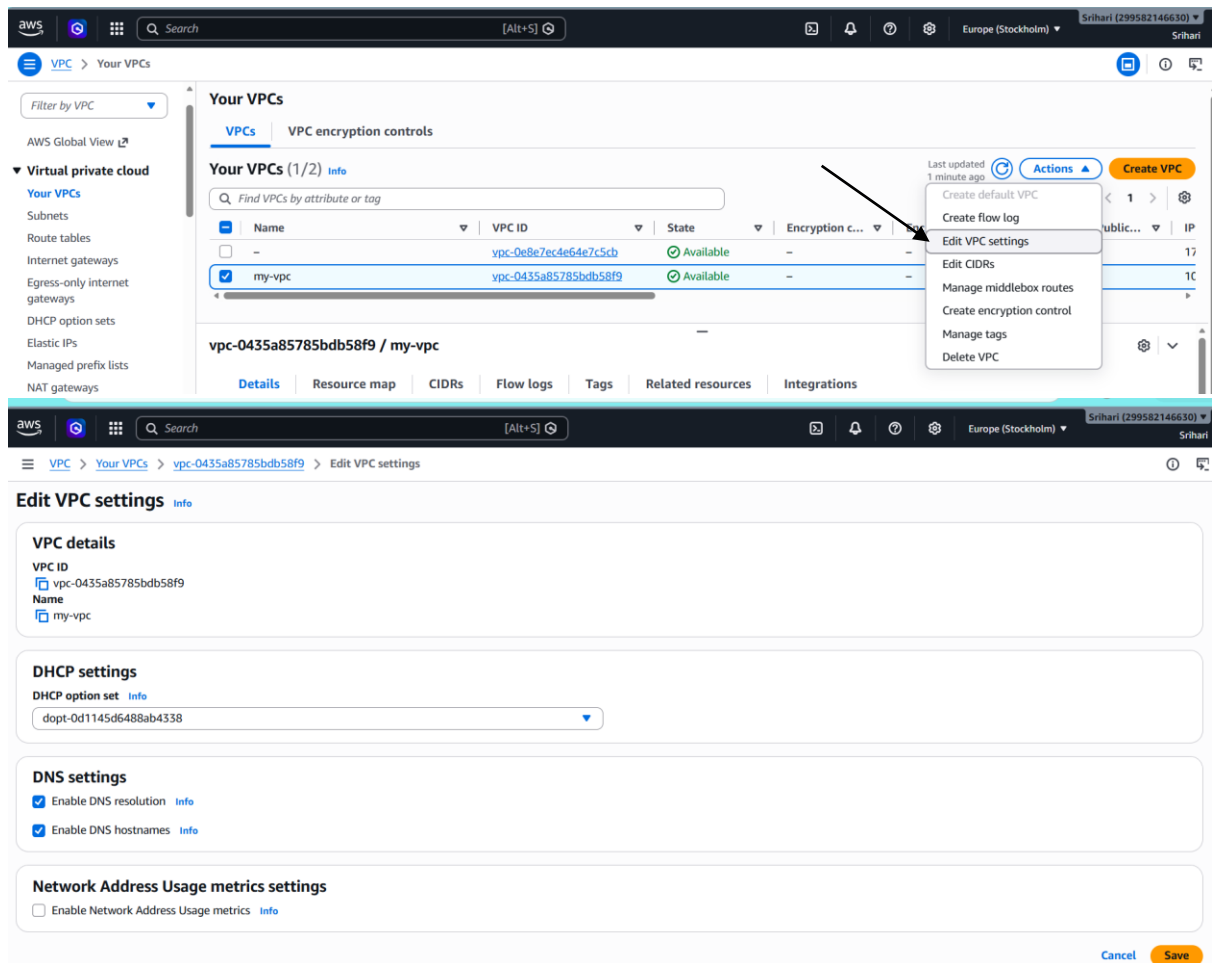
eu-north-1b

- Public-Subnet-2
- Private-Subnet-1

Route tables (1)
Route network traffic to resources
rtb-09780b330db1d3aba

2. Enable DNS Hostname in VPC.

- Go to the **AWS Console** → **VPC** service.
- In the left menu, click **Your VPCs**.
- Select your VPC (example: my-vpc-01).
- At the bottom, go to the **Details** tab.
- Click **Edit VPC settings**.
- Enable the checkbox:
DNS hostnames
- Ensure **DNS resolution** is also enabled.
- Click **Save changes**.



3. Enable Auto Assign Public IP in 2 public subnets.

Go to the VPC Console

- Open AWS Console → VPC → Subnets.

Select Public Subnet 1

- Click on your subnet:
Public-subnet-01
- Choose the Subnet actions dropdown.
- Click Edit subnet settings.

Enable Auto-Assign Public IPv4

- Tick the checkbox:
✓ Enable auto-assign public IPv4 address
- Click Save.

Repeat for Public Subnet 2

- Select your second subnet:
Public-subnet-02
- Again click:
 - Edit subnet settings
 - Enable Auto-assign public IPv4
 - Save.

The screenshot shows the AWS Management Console interface for the 'Subnets' page. The left sidebar contains navigation links for 'Virtual private cloud', 'Security', and 'PrivateLink and Lattice'. The main content area displays a table of subnets. The 'Public-Subnet-1' is selected, and the 'Edit subnet settings' dropdown menu is open, showing options like 'View details', 'Create flow log', 'Edit subnet settings', 'Edit IPv6 CIDRs', 'Edit network ACL association', 'Edit route table association', 'Edit CIDR reservations', 'Share subnet', 'Manage tags', and 'Delete subnet'. The 'Auto-assign IP settings' section is visible, showing 'Enable auto-assign public IPv4 address' checked. The 'Resource-based name (RBN) settings' section is also visible, showing 'Enable resource name DNS A record on launch' checked.

Name	Subnet ID	State	VPC
Public-Subnet-2	subnet-0dfc720a9aa1b18b4	Available	vpc-0435a85785
Public-Subnet-1	subnet-050f61ea23f7fd1d9	Available	vpc-0435a85785
private-Subnet-2	subnet-0f6a4d4d00324bafa	Available	vpc-0435a85785
Private-Subnet-1	subnet-0e927b6727d4c5570	Available	vpc-0435a85785
-	subnet-079d3e04543e35fe9	Available	vpc-0e8e7ec4e6
-	subnet-07f5ff49850fb8795	Available	vpc-0e8e7ec4e6

subnet-050f61ea23f7fd1d9 / Public-Subnet-1

Details

Subnet ID	Subnet ARN	State	Block Public Access
subnet-050f61ea23f7fd1d9	arn:aws:ec2:eu-north-1:299582146630:subnet/subnet-050f61ea23f7fd1d9	Available	Off

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address

☐ Enable auto-assign customer-owned IPv4 address

Resource-based name (RBN) settings

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

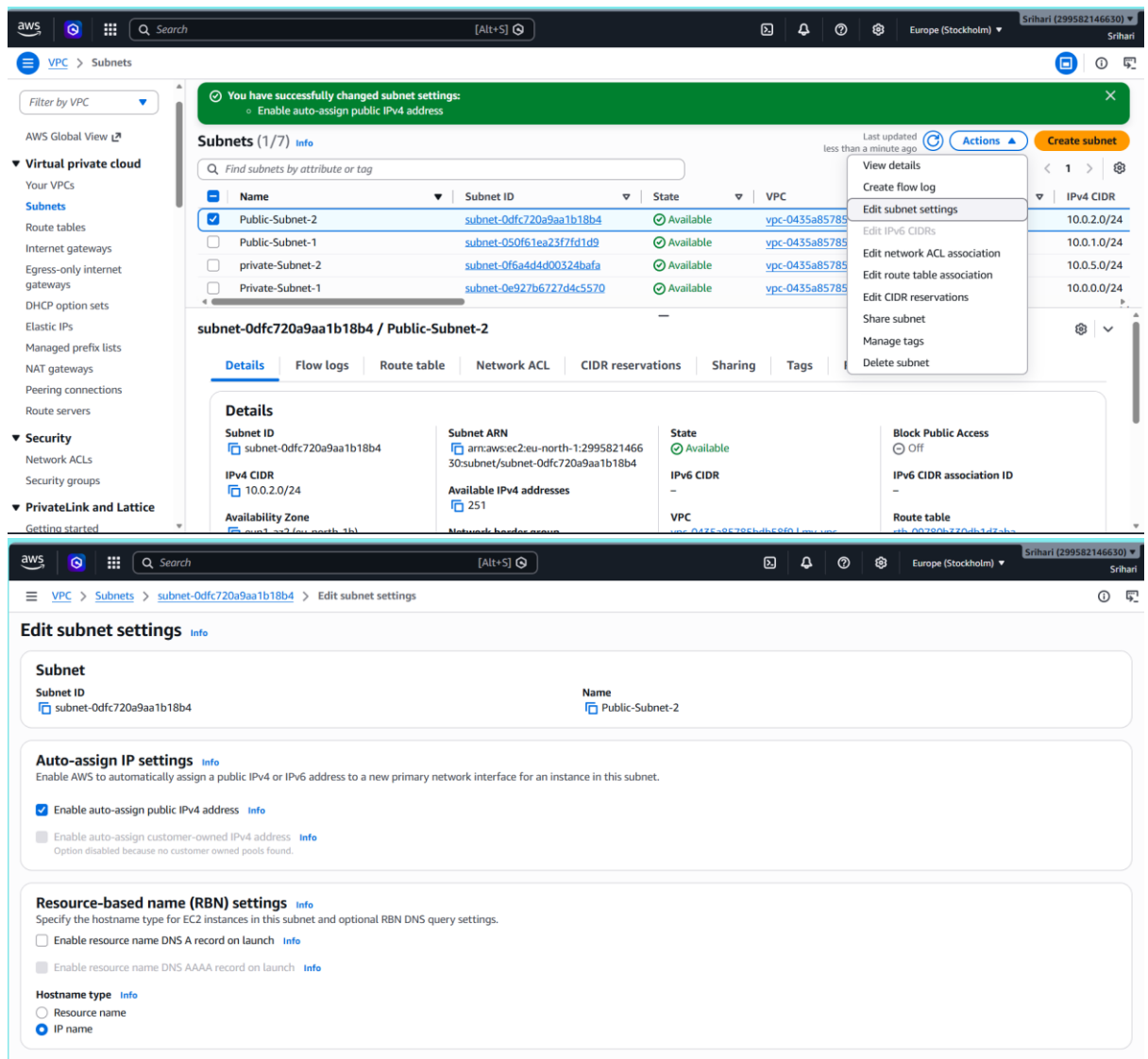
☐ Enable resource name DNS A record on launch

☐ Enable resource name DNS AAAA record on launch

Hostname type

☐ Resource name

☒ IP name



4. Add 2 private subnets in private route table.

Step 1: Go to Route Tables

- Open AWS Console → VPC → Route Tables.

Step 2: Select Your Route Table

Step 3: Open Subnet Associations

- In the bottom panel → select Subnet associations tab.
- Click Edit subnet associations.

Step 4: Select Both Private Subnets

Tick the checkboxes for:

- Private-subnet-01
- Private-subnet-02

Step 5: Save

- Click Save associations.

Now both private subnets are attached to the Private Route Table

The image shows two screenshots from the AWS Management Console. The top screenshot displays the 'Route tables' page for the 'vpc-0435a85785bdb58f9' VPC. It lists two route tables: 'rtb-05263edb131b81b95' and 'rtb-09780b330db1d3aba'. The second route table is selected, and its details are shown below, including its ID, VPC, and owner information. The bottom screenshot shows the 'Edit subnet associations' page for the selected route table 'rtb-09780b330db1d3aba'. It lists available subnets: 'Public-Subnet-1', 'Public-Subnet-2', 'Private-Subnet-1', and 'private-Subnet-2'. The two private subnets are selected, and they are listed in the 'Selected subnets' section at the bottom. The 'Save associations' button is visible in the bottom right corner.

Route tables (1/2) Info

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-05263edb131b81b95	-	-	Yes	vpc-0e8e7ec4e64e7c5cb
-	rtb-09780b330db1d3aba	-	-	Yes	vpc-0435a85785bdb58f9 my-

rtb-09780b330db1d3aba

Details

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-09780b330db1d3aba	Yes	-	-

VPC
vpc-0435a85785bdb58f9 | my-vpc

Owner ID
299582146630

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public-Subnet-1	subnet-050f61ea23f7fd1d9	10.0.1.0/24	-	Main (rtb-09780b330db1d3aba)
Public-Subnet-2	subnet-0dfc720a9aa1b18b4	10.0.2.0/24	-	Main (rtb-09780b330db1d3aba)
Private-Subnet-1	subnet-0e927b6727d4c5570	10.0.0.0/24	-	Main (rtb-09780b330db1d3aba)
private-Subnet-2	subnet-0f6a4d4d00324bafa	10.0.5.0/24	-	Main (rtb-09780b330db1d3aba)

Selected subnets

subnet-0e927b6727d4c5570 / Private-Subnet-1

subnet-0f6a4d4d00324bafa / private-Subnet-2

Cancel **Save associations**

5. Add 2 public subnets in public route table.

Step 1: Go to Route Tables

- Open AWS Console → VPC → Route Tables.

Step 2: Select Your Route Table

Step 3: Open Subnet Associations

- In the bottom panel → select Subnet associations tab.
- Click Edit subnet associations.

Step 4: Select Both Public Subnets

Tick the checkboxes for:

- Public-subnet-01
- Public-subnet-02

Step 5: Save

- Click Save associations.

The first screenshot shows the 'Route tables' page in the AWS VPC console. A green notification banner at the top states: 'You have successfully updated subnet associations for rtb-09780b330db1d3aba.' The 'Route tables (1/2)' table lists two route tables. The second route table, 'rtb-09780b330db1d3aba', is selected and its details are shown below. It lists two private subnets: 'Private-Subnet-1' (subnet-0e927b6727d4c5570) and 'private-Subnet-2' (subnet-0f6a4d4d00324bafa). Below this, a section titled 'Subnets without explicit associations (2)' lists two public subnets: 'Public-Subnet-1' (subnet-050f61ea23f7fd1d9) and 'Public-Subnet-2' (subnet-0dfc720a9aa1b18b4).

The second screenshot shows the 'Edit subnet associations' page for the selected route table. It displays a table of 'Available subnets (2/4)'. The first two subnets, 'Public-Subnet-1' and 'Public-Subnet-2', are selected with checkboxes. Below the table, the 'Selected subnets' section shows the selected subnets: 'subnet-0dfc720a9aa1b18b4 / Public-Subnet-2' and 'subnet-050f61ea23f7fd1d9 / Public-Subnet-1'. At the bottom right, there are 'Cancel' and 'Save associations' buttons.

6. Public route table will have the routes to internet and local.

- Go to VPC Console → Internet Gateways
- Click Create Internet Gateway
- Name: MyIGW
- After creating → Select it
- Click Actions → Attach to VPC
- Choose your VPC (my-vpc-01) → Attach

- VPC → Route Tables → public-rt → Routes → Edit routes
- Add:

Destination Target
 0.0.0.0/0 Select Internet Gateway

The first screenshot shows the 'Internet gateways' page in the AWS Management Console. It displays a table with one entry: an Internet Gateway with ID 'igw-0af361ac47cb431bf' in an 'Attached' state, associated with VPC ID 'vpc-0e8e7ec4e64e7c5cb'. The left sidebar shows the navigation menu with 'Internet gateways' selected under 'Virtual private cloud'.

The second screenshot shows the details page for the Internet Gateway 'igw-04021612de8fc848a / MyIGW'. It shows the gateway is in a 'Detached' state. A green notification banner at the top states: 'The following internet gateway was created: igw-04021612de8fc848a - MyIGW. You can now attach to a VPC to enable the VPC to communicate with the internet.' The 'Actions' menu includes 'Attach to VPC', 'Detach from VPC', 'Manage tags', and 'Delete'. The 'Tags' section shows a single tag with the key 'Name' and value 'MyIGW'.

The third screenshot shows the 'Attach to VPC (igw-04021612de8fc848a)' dialog box. It prompts the user to 'Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.' Under 'Available VPCs', a search bar contains 'vpc-0435a85785bdb58f9'. Below the search bar, a dropdown menu shows the selected VPC: 'vpc-0435a85785bdb58f9 - my-vpc'. At the bottom right, there are 'Cancel' and 'Attach internet gateway' buttons.

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC

Internet gateways

igw-04021612de8fc848a

Internet gateway igw-04021612de8fc848a successfully attached to vpc-0435a85785bdb58f9

igw-04021612de8fc848a / MyIGW

Actions

Details

Internet gateway ID

igw-04021612de8fc848a

State

Attached

VPC ID

vpc-0435a85785bdb58f9 | my-vpc

Owner

299582146630

Tags (1)

Search tags

Manage tags

< 1 >

Key

Value

Name

MyIGW

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC

Route tables

Route tables (1/2)

Find route tables by attribute or tag

Last updated 11 minutes ago

Actions

Create route table

< 1 >

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-05263edb131b81b95	-	-	Yes	vpc-0e8e7ec4e64e7c5cb
☒	-	rtb-09780b330db1d3aba	2 subnets	-	Yes	vpc-0435a85785bdb58f9 my-

rtb-09780b330db1d3aba

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (1)

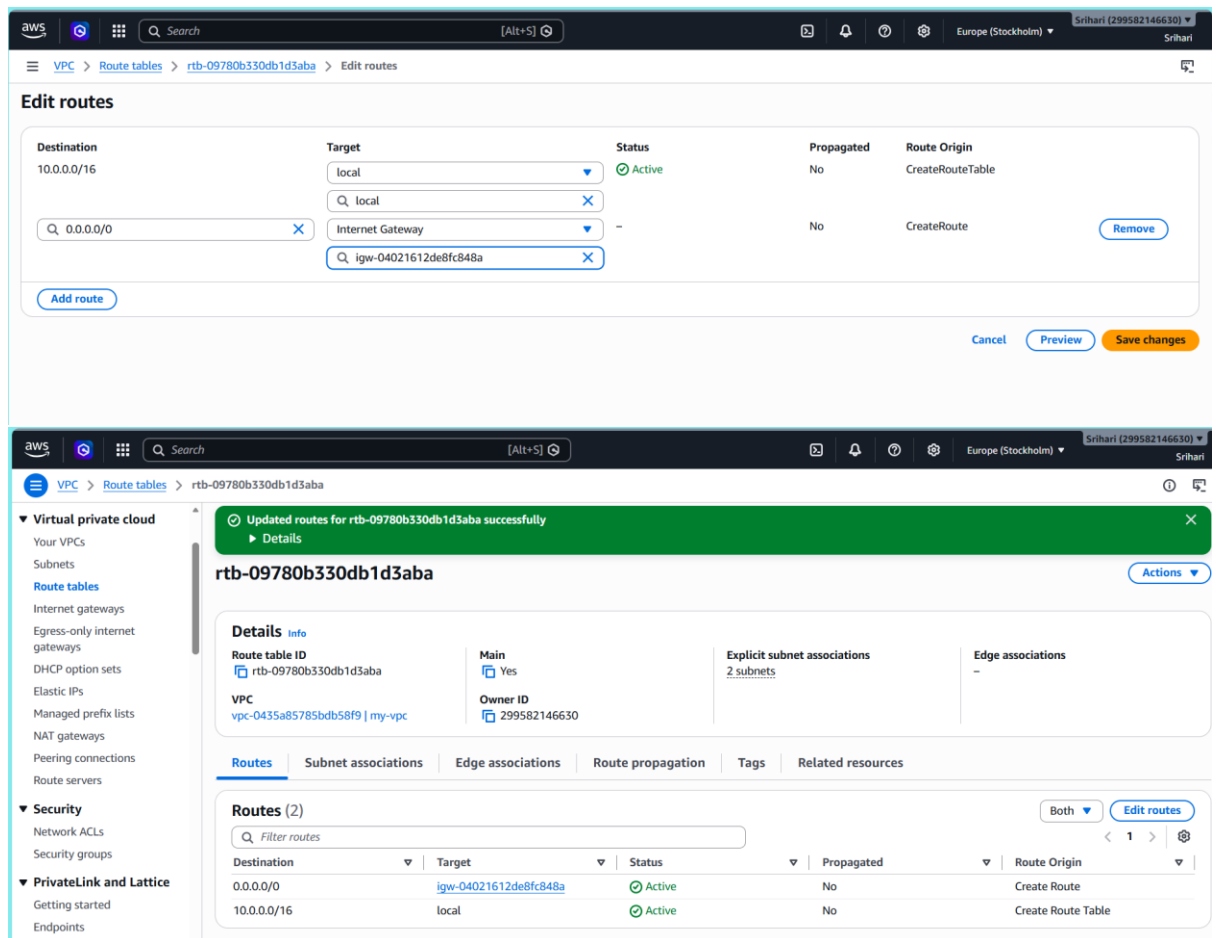
Filter routes

Both

Edit routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table



7. Create EC2 in public subnet with t2.micro and install PHP.

- Open the AWS console in your browser and launch a new EC2 instance named **php-server**.
- Select the **Amazon Linux** AMI
- Under **Network Settings**, choose your **VPC** and select the **public subnet**.
- In the **Security Group**, add the following inbound rules:
- **SSH (port 22)**
- **HTTP (port 80)**
- Launch the instance and connect to it using SSH.
- Run:
 - `sudo yum update -y`
 - `sudo yum install httpd -y`
 - `systemctl status httpd`
 - `sudo systemctl start httpd`
 - `sudo yum install -y php php-cli php-mysqlnd`
 - `sudo systemctl restart httpd`
 - then with public ip check on browser

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

EC2

Security Groups

sg-06fcd18c73724a36e

Edit inbound rules

Edit inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Info

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sg-0a4c0e1d68cb2de90	All traffic	All	All	Custom		Delete
-	SSH	TCP	22	Anywh...	sg-06fcd18c73724a36e	Delete
-	HTTP	TCP	80	Anywh...	0.0.0.0/0	Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

```
[ec2-user@ip-10-0-1-70 ~]$ sudo su
[root@ip-10-0-1-70 ec2-user]# sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-1-70 ec2-user]# sudo yum install httpd -y
Last metadata expiration check: 0:00:18 ago on Wed Jul 15 14:40:36 2026.
Dependencies resolved.

Package                               Architecture      Version            Repository
-----
Installing:
httpd                                x86_64            2.4.68-1.amzn2023.0.1  amazonlinux
Installing dependencies:
apr                                x86_64            1.7.5-1.amzn2023.0.4  amazonlinux
apr-util                            x86_64            1.6.3-1.amzn2023.0.2  amazonlinux
apr-util-ldap                       x86_64            1.6.3-1.amzn2023.0.2  amazonlinux
generic-logos-httpd                 x86_64            18.0.0-12.amzn2023.0.3  amazonlinux
httpd-core                           x86_64            2.4.68-1.amzn2023.0.1  amazonlinux
httpd-filesystem                     x86_64            2.4.68-1.amzn2023.0.1  amazonlinux
httpd-tools                          x86_64            2.4.68-1.amzn2023.0.1  amazonlinux
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2  amazonlinux
mailcap                              x86_64            2.1.49-3.amzn2023.0.3  amazonlinux
Installing weak dependencies:
apr-util-openssl                     x86_64            1.6.3-1.amzn2023.0.2  amazonlinux
mod_http2                            x86_64            2.0.42-1.amzn2023.0.1  amazonlinux
mod_lua                              x86_64            2.4.68-1.amzn2023.0.1  amazonlinux

Transaction Summary
Install 13 Packages

[ec2-user@ip-10-0-1-70 ec2-user]# sudo systemctl start httpd
sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-1-70 ec2-user]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-07-15 14:41:08 UTC; 16s ago
     Docs: man:httd.service(8)
  Main PID: 23576 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1059)
   Memory: 13.9M
      CPU: 85ms
    CGroup: /system.slice/httpd.service
            └─23576 /usr/sbin/httpd -DFOREGROUND
            └─23669 /usr/sbin/httpd -DFOREGROUND
            └─23676 /usr/sbin/httpd -DFOREGROUND
            └─23728 /usr/sbin/httpd -DFOREGROUND
            └─23735 /usr/sbin/httpd -DFOREGROUND

Jul 15 14:41:08 ip-10-0-1-70.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jul 15 14:41:08 ip-10-0-1-70.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jul 15 14:41:08 ip-10-0-1-70.eu-north-1.compute.internal httpd[23576]: Server configured, listening on: port 80
[ec2-user@ip-10-0-1-70 ec2-user]#

[ec2-user@ip-10-0-1-70 ec2-user]# sudo dnf install php php-cli php-fpm -y
Last metadata expiration check: 0:03:08 ago on Wed Jul 15 14:40:36 2026.
Dependencies resolved.

Package                               Architecture      Version            Repository
-----
Installing:
php8.5                               x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
php8.5-fpm                           x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
Installing dependencies:
libsodium                             x86_64            1.0.19-5.amzn2023    amazonlinux
libxml2                               x86_64            1.1.43-3.amzn2023.0.3  amazonlinux
nghttp-filesystem                     x86_64            1.1.30.3-1.amzn2023.0.1  amazonlinux
php8.5-cli                            x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
php8.5-common                         x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
php8.5-process                        x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
php8.5-xml                            x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
Installing weak dependencies:
php8.5-mbstring                       x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
php8.5-pdo                            x86_64            8.5.7-1.amzn2023.0.1  amazonlinux
php8.5-sodium                         x86_64            8.5.7-1.amzn2023.0.1  amazonlinux

Transaction Summary
Install 12 Packages

Total download size: 11 M
Installed size: 60 M

[ec2-user@ip-10-0-1-70 ec2-user]# php -v
PHP 8.5.7 (cli) (built: Jun 2 2026 20:59:56) (NTS gcc x86_64)
Copyright (C) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.7, Copyright (C) Zend Technologies
    with Zend OPcache v8.5.7, Copyright (C), by Zend Technologies
[ec2-user@ip-10-0-1-70 ec2-user]# sudo systemctl restart httpd
[ec2-user@ip-10-0-1-70 ec2-user]# sudo vi /var/www/html/info.php
[ec2-user@ip-10-0-1-70 ec2-user]# sudo vi /var/www/html/info.php
```

The image shows a terminal window at the top and a web browser window at the bottom. The terminal window has a dark background and shows the command `phpinfo();` being executed. The web browser window shows the output of the `phpinfo()` command, displaying various system and PHP configuration details.

Terminal Window:

```
root@ip-10-0-1-70:/home/ec2-user
?php
phpinfo();
?>
```

Web Browser Window:

Not secure 13.61.155.7/info.php

PHP Version 8.5.7

System	Linux ip-10-0-1-70-eu-north-1-compute-internal 6.18.38-69.138.amzn2023.x86_64 #2 SMP PREEMPT_DYNAMIC Thu Jul 9 21:06:10 UTC 2020 x86_64
Build Date	Jun 2 2026 20:59:56
Build System	Linux
Build Provider	Amazon Linux
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dbm.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-posix.ini, /etc/php.d/20-shmop.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sodium.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-sysmsg.ini, /etc/php.d/20-system.ini, /etc/php.d/20-sysvshm.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmireader.ini
PHP API	20250925
PHP Extension	20250925
Zend Extension	40250925
Zend Extension Build	API20250925.NTS
PHP Extension Build	API20250925.NTS
PHP Integer Size	64 bits
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring
Zend Max Execution Timers	disabled

8. Configure NAT gateway in public subnet and connect to private instance.

EC2

Resources

You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:

Instances (running)	0	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	5
Key pairs	4	Load balancers	0	Placement groups	0
Security groups	12	Snapshots	0	Volumes	5

Launch instance

To get started, launch an Amazon EC2 Instance, which is a virtual server in the cloud.

[Launch instance](#)

[Migrate a server](#)

Service health

[AWS Health Dashboard](#)

Region: Europe (Stockholm)

Status: This service is operating normally.

EC2 cost

Date range: Past 6 months

Region: Global

Costs in your free plan account are covered by credits.

Credits remaining: \$116.66 USD

Days remaining: 160 (December 22, 2026)

Unable to load

[Analyze your costs in Cost Explorer](#)

Account attributes

[Default VPC](#)

Elastic IP addresses

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Ass
No Elastic IP addresses found in this Region					

[Allocate Elastic IP address](#)

Elastic IP addresses

Elastic IP address allocated successfully. Elastic IP address 13.49.242.240

[Associate this Elastic IP address](#)

Elastic IP addresses (1)

Find elastic IP addresses by attribute or tag

Public IPv4 address: 13.49.242.240

[Clear filters](#)

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Ass
-	13.49.242.240	Public IP	eipalloc-09eab5a5b011dbfee	-	-

[Allocate Elastic IP address](#)

VPC dashboard

Filter by VPC

NAT gateways

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID
No NAT gateways found						

[Create NAT gateway](#)

Select a NAT gateway

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC

NAT gateways

Create NAT gateway

Create NAT gateway

Info

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

My-NAT-Gateway

The name can be up to 256 characters long.

Availability mode

Info

Choose whether to deploy across all zones in the region or restrict to a single availability zone.

Regional - new

Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

Zonal

Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC

Select a VPC in which to create the regional NAT gateway.

vpc-0435a85785bdb58f9 (my-vpc)

Connectivity type

Select a connectivity type for the NAT gateway.

Public

Private

Method of Elastic IP (EIP) allocation

Info

Choose how IP addresses are associated with NAT gateways.

Automatic

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC

NAT gateways

nat-15c43dad4b3b77966

NAT gateway nat-15c43dad4b3b77966 | My-NAT-Gateway was created successfully.

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

nat-15c43dad4b3b77966 / My-NAT-Gateway

Actions

Details

NAT gateway ID

nat-15c43dad4b3b77966

NAT gateway ARN

arn:aws:ec2:eu-north-1:29958214663:0:natgateway/nat-15c43dad4b3b77966

VPC

vpc-0435a85785bdb58f9 / my-vpc

Availability mode

Regional

Connectivity type

Public

Method of EIP allocation

Automatic

State

Pending

Created

Wednesday, July 15, 2026 at 20:47:02 GMT+5:30

State message

Info

-

Deleted

-

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC

Route tables

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Route tables (1/2)

Info

Last updated 10 minutes ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	-	rtb-09780b330db1d3aba	2 subnets	-	Yes	vpc-0435a85785bdb58f9 my-
☐	-	rtb-05263edb131b81b95	-	-	Yes	vpc-0e8e7ec4e64e7c5cb

rtb-09780b330db1d3aba

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC

Route tables

Create route table

Create route table

Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

my-route-table-01

VPC

The VPC to use for this route table.

vpc-0435a85785bdb58f9 (my-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q my-route-table-01

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC > Route tables > rtb-01047188bc2ae46e0 > Edit routes

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Q 0.0.0.0/0

X

Q local

X

NAT Gateway

X

Q nat-15c43dad4b3b77966

X

Add route

Cancel Preview Save changes

aws

Search

[Alt+S]

Europe (Stockholm)

Srihari (299582146630)

VPC > Route tables > rtb-01047188bc2ae46e0 > Edit subnet associations

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	Public-Subnet-2	subnet-0dfc720a9aa1b18b4	10.0.2.0/24	-	rtb-09780b330db1d3aba
☒	private-Subnet-2	subnet-0f6a4d4d00324bafa	10.0.5.0/24	-	Main (rtb-09780b330db1d3aba)
☐	Public-Subnet-1	subnet-050f61ea23f7fd1d9	10.0.1.0/24	-	rtb-09780b330db1d3aba
☒	Private-Subnet-1	subnet-0e927b6727d4c5570	10.0.0.0/24	-	Main (rtb-09780b330db1d3aba)

Selected subnets

subnet-0f6a4d4d00324bafa / private-Subnet-2

subnet-0e927b6727d4c5570 / Private-Subnet-1

Cancel Save associations

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

awskey

Create new key pair

▼ Network settings Info

VPC - required Info

vpc-0435a85785bdb58f9 (my-vpc)

10.0.0.0/16

Subnet Info

subnet-0f6a4d4d00324bafa

private-Subnet-2

VPC: vpc-0435a85785bdb58f9 Owner: 299582146630
Availability Zone: eu-north-1a (eu1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.5.0/24

Create new subnet

Auto-assign public IP Info

Enable

Firewall (security groups) Info

Create security group

Select existing security group

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

▼ Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...read more

ami-068b5bc67e48209c1

Virtual server type (instance type)

t3.micro

Firewall (security group)

default

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

- **Switch to Root User:** The user switches to the root user (`sudo su -`).
- **Install Java:** The command `yum install java-11-amazon-corretto` is used to install Amazon Corretto 11 (an OpenJDK distribution). The process shows dependencies being resolved and the transaction summary for the installation.

2. Download and Extract Apache Tomcat

- **Navigate to /opt:** The user changes directory to /opt (`cd /opt`).
- **Download Tomcat:** The command `wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.112/bin/apache-tomcat-9.0.112.tar.gz` downloads the Tomcat 9.0.112 archive.
- **Extract Tomcat:** The command `tar xvf apache-tomcat-9.0.112.tar.gz` extracts the contents of the archive, creating the `apache-tomcat-9.0.112` directory.

3. Start Tomcat and Check Default Page

- **Navigate to Tomcat bin:** The user changes directory to the Tomcat binary directory (`cd apache-tomcat-9.0.112/bin`).
- **Start Tomcat:** The command `./startup.sh` starts the Apache Tomcat server. The output confirms that Tomcat has started.
- **Check Tomcat Default Page:**
 - The user navigates to the webapps directory (`cd ../webapps`).
 - The command `curl http://172.168.0.139:8080/` is used to check the default Tomcat welcome page on port 8080.
 - The output shows the HTML content of the Tomcat welcome page, confirming the server is running and accessible.

4. Deploy Sample Application (sample.war)

- **Navigate to webapps:** The user is in the Tomcat webapps directory.
- **Download Sample WAR:** The command `wget https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war` downloads the standard Tomcat sample web application archive (`sample.war`).
- **Deployment:** Tomcat automatically deploys the `.war` file, creating a new directory named `sample` inside `webapps`.
- **Verification of Deployment:** The `ls` command shows the newly deployed sample directory along with `sample.war`. The user then checks the contents of the sample directory.

5. Access the Deployed Sample Application

- **Access Sample Application:** The command `curl http://172.168.0.139:8080/sample` is executed to access the newly deployed sample application.
- The output of the `curl` command (not fully visible, but implied by the command itself) would show the HTML content returned by the sample application, confirming successful deployment and access.

vpc-flowlogs-srihari Info

< **Objects** Metadata Properties Permissions Metrics Management File system >

Objects (0)

[Refresh](#)
[Copy S3 URI](#)
[Copy URL](#)
[Download](#)
[Open](#)
[Delete](#)
[Actions](#)

[Create folder](#)
[Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix < 1 > [Settings](#)

☐ Name
 ☐ Type
 ☐ Last modified
 ☐ Size
 ☐ Storage class

No objects

You don't have any objects in this bucket.

[Upload](#)

aws

Search

Europe (Stockholm)

Srihari (299582146630)

VPC > Your VPCs > fl-0c61ff323119e1f00

Successfully created flow log for the following resource:

vpc-0435a85785bdb58f9

[Explore integration options](#)

fl-0c61ff323119e1f00 / vpc-flowlogs-srihari

[Actions](#)

Details

Flow Log ID fl-0c61ff323119e1f00	Destination Type s3	Traffic Type All	File Format Plain text
Name vpc-flowlogs-srihari	Destination Name vpc-flowlogs-srihari	Max Aggregation Interval 1 minute	Hive Compatible Partitions Not enabled
State Active	IAM Role -	Log Format Default	Partition Logs Daily
Creation Time Thursday, July 16, 2026 at 16:47:22 GMT+5:30	Cross Account IAM Role -	Tag field specifications -	

```
[root@ip-10-0-1-120 ec2-user]# ping google.com
PING google.com (216.58.201.238) 56(84) bytes of data:
64 bytes from par10b33-in-f14.1e100.net (216.58.201.238): icmp_seq=1 ttl=119 time=4.00 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=2 ttl=119 time=3.99 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=3 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=4 ttl=119 time=3.96 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=5 ttl=119 time=3.96 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=6 ttl=119 time=3.97 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=7 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=8 ttl=119 time=3.97 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=9 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=10 ttl=119 time=3.97 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=11 ttl=119 time=3.99 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=12 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=13 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=14 ttl=119 time=3.97 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=15 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=16 ttl=119 time=4.00 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=17 ttl=119 time=4.00 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=18 ttl=119 time=3.97 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=19 ttl=119 time=4.06 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=20 ttl=119 time=3.99 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=21 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=22 ttl=119 time=3.99 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=23 ttl=119 time=3.99 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=24 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=25 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=26 ttl=119 time=4.00 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=27 ttl=119 time=4.00 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=28 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=29 ttl=119 time=3.97 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=30 ttl=119 time=3.98 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=31 ttl=119 time=4.00 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=32 ttl=119 time=4.30 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=33 ttl=119 time=4.37 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=34 ttl=119 time=3.99 ms
64 bytes from lcarbn-af-in-f14.1e100.net (216.58.201.238): icmp_seq=35 ttl=119 time=4.01 ms
```

Amazon S3

Buckets

vpc-flowlogs-srihari

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

vpc-flowlogs-srihari

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

☐

Name

☐

AWSLogs/

Type

Folder

Last modified

-

Size

-

Storage class

-

Europe (Stockholm)

Srihari (299582146630)

Srihari

Amazon S3

Buckets

vpc-flowlogs-srihari

AWSLogs/

299582146630/

vpcflowlogs/

eu-north-1/

2026/

07/

16/

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Account and organization settings

16/

Copy S3 URI

Objects

Properties

Objects (4)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

☐

Name

☐

299582146630_vpcflowlogs_eu-north-1_fl-0c61ff323119e1f00_20260716T1115Z_913208bc.log.gz

Type

gz

Last modified

July 16, 2026, 16:51:40 (UTC+05:30)

Size

910.0 B

Storage class

Standard

☐

299582146630_vpcflowlogs_eu-north-1_fl-0c61ff323119e1f00_20260716T1120Z_669e4b80.log.gz

Type

gz

Last modified

July 16, 2026, 16:51:40 (UTC+05:30)

Size

516.0 B

Storage class

Standard

☐

299582146630_vpcflowlogs_eu-north-1_fl-0c61ff323119e1f00_202607

Type

gz

Last modified

July 16, 2026, 16:56:40 (UTC+05:30)

Size

2.4 KB

Storage class

Standard

Europe (Stockholm)

Srihari (299582146630)

Srihari

VPC

Your VPCs

fl-038c9b16ac8b64107

VPC dashboard

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Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Successfully created flow log for the following resource:

vpc-0435a85785bdb58f9

fl-038c9b16ac8b64107 / vpc flow logs

Actions

Details

Flow Log ID

fl-038c9b16ac8b64107

Name

vpc flow logs

State

Active

Creation Time

Thursday, July 16, 2026 at 17:30:36 GMT+5:30

Destination Type

cloud-watch-logs

Destination Name

VPCFlowlogs

IAM Role

arn:aws:iam::299582146630:role/service-role/VPCFlowLogs-Cloudwatch-1784203067866

Cross Account IAM Role

-

Traffic Type

All

Max Aggregation Interval

1 minute

Log Format

Default

Tag field specifications

-

File Format

-

Hive Compatible Partitions

-

Partition Logs

-

Tags

Integrations

Log groups

Data sources - new

Summary

Get-set-go with unified data store

Log groups (0)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search

Exact match

Log group

Log class

Anomaly d...

Deletion p...

Bearer tok...

Data pr...

Sensitiv...

Retention

Metric ...

There are no log groups.

CloudWatch

Log events

Log Management

Log Analytics

Log Anomalies

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

Timestamp	Message
2026-07-16T12:09:22.000Z	2 299582146630 - 45.205.1.247 16.192.110.113 50964 4444 6 1 40 1784203762 1784203787 ACCEPT OK
2026-07-16T12:09:22.000Z	2 299582146630 - 85.217.140.51 16.192.110.113 36200 56683 6 1 52 1784203762 1784203787 ACCEPT OK
2026-07-16T12:09:23.000Z	2 299582146630 - 35.203.210.199 13.63.119.210 49946 4083 6 1 44 1784203763 1784203763 ACCEPT OK
2026-07-16T12:09:25.000Z	2 299582146630 - - - - - 1784203765 1784203796 - NODATA
2026-07-16T12:09:26.000Z	2 299582146630 - - - - - 1784203766 1784203766 - NODATA
2026-07-16T12:09:26.000Z	2 299582146630 - - - - - 1784203766 1784203766 - SKIPDATA
2026-07-16T12:09:26.000Z	2 299582146630 - 35.203.210.212 13.63.119.210 52449 2907 6 1 44 1784203766 1784203767 ACCEPT OK
2026-07-16T12:09:38.000Z	2 299582146630 - 5.181.21.156 16.192.110.113 51489 8900 6 1 40 1784203778 1784203778 ACCEPT OK
2026-07-16T12:09:38.000Z	2 299582146630 - - - - - 1784203778 1784203778 - NODATA
2026-07-16T12:09:38.000Z	2 299582146630 - - - - - 1784203778 1784203778 - SKIPDATA
2026-07-16T12:09:57.000Z	2 299582146630 - 85.217.149.66 16.192.110.113 55170 21270 6 1 52 1784203797 1784203809 ACCEPT OK

Back to top